

1 The SS–RS–GD inequalities for shuffled SGD

The finite-sum objective $\min_z \frac{1}{n} \sum_{i=1}^n f_i(z)$ at the heart of empirical risk minimization is optimized in practice by one of three schemes that differ only in the order they sweep the n components each epoch. *Gradient descent* (GD) uses the full batch at every step; RANDOMSHUFFLE SGD draws a fresh random permutation of the components every epoch; and SINGLESHUFFLE SGD draws a single permutation at the outset and reuses it for all K epochs. These variants have by now an extensive theory [3, 6, 7, 1, 5, 9]. Yun, Sra, and Jadbabaie [10] set out to make the ordering precise as a COLT open problem. Writing $P_\sigma := A_{\sigma(1)} A_{\sigma(2)} \cdots A_{\sigma(n)}$ for the ordered product of symmetric matrices $A_1, \dots, A_n$ along a permutation $\sigma \in \mathfrak{S}_n$, the expected iterate after K epochs of the three schemes is governed by the operators

$$W_{\text{SS}} := \frac{1}{n!} \sum_{\sigma \in \mathfrak{S}_n} P_\sigma^K, \quad W_{\text{RS}} := \left(\frac{1}{n!} \sum_{\sigma \in \mathfrak{S}_n} P_\sigma \right)^K, \quad W_{\text{GD}} := \left(\frac{1}{n} \sum_{i=1}^n A_i \right)^{nK} \quad (1.1)$$

Note single-shuffle reuses one permutation for all K epochs, so its K -th power sits *inside* the average; reshuffling draws a fresh permutation each epoch, with the power *outside*; and W_{GD} is full-batch gradient descent. Their conjecture is the following.

Conjecture 1.1 (Yun–Sra–Jadbabaie [10], SS–RS–GD inequalities). For every $n \geq 2$, $K \geq 1$, and $d \geq 1$ there exists a constant $\eta_{n,K} \in (0, 1]$ such that the following holds. If real symmetric $d \times d$ matrices $A_1, \dots, A_n$ satisfy $(1 - \eta_{n,K})I \preceq A_i \preceq I$ for all i , then

$$\|W_{\text{SS}}\| \leq \|W_{\text{RS}}\| \leq \|W_{\text{GD}}\|. \quad (1.2)$$

The conjecture requires the matrices to be well-conditioned; this is necessary as the RS–GD inequality for general PSD matrices (originally conjectured by Recht and Ré [8]) is *false* for all $n \geq 5$ [4, 2].

Our results. The COLT open question is resolved, in particular, the SS–RS inequality fails while the RS–GD inequality stands.

Theorem 1.2 (Refutation; SS–RS inequality fails). *Conjecture 1.1 is false. Concretely, fix $n = 3$ and $K = 2$. For every candidate $\eta_{3,2} \in (0, 1]$ there exist real symmetric positive definite 4×4 matrices A_1, A_2, A_3 with $(1 - \eta_{3,2})I \preceq A_i \preceq I$ for which*

$$\|W_{\text{SS}}\| > \|W_{\text{RS}}\|.$$

Theorem 1.3 (RS–GD inequality holds). *For every $n \geq 2$, every $K \geq 1$, every $d \geq 1$, and every real symmetric $A_1, \dots, A_n$ with*

$$\left(1 - \frac{1}{4n^2 + 1}\right)I \preceq A_i \preceq I \quad (1 \leq i \leq n),$$

one has $\|W_{\text{RS}}\| \leq \|W_{\text{GD}}\|$. Thus the second inequality of (1.2) holds with the constant $\eta_{n,K}^{\text{RS-GD}} = \frac{1}{4n^2 + 1}$.

1.1 Setup and notation

Notation I is the identity (of the dimension clear from context), $\preceq$ is the Löwner order, and $\|\cdot\|$ is the spectral norm (largest singular value), which equals $\max_i |\lambda_i(\cdot)|$ for symmetric matrices. Recall from (1.1) the products $P_\sigma = A_{\sigma(1)}A_{\sigma(2)}\cdots A_{\sigma(n)}$ and the operators $W_{\text{SS}}, W_{\text{RS}}, W_{\text{GD}}$. It is convenient to set

$$R := \frac{1}{n!} \sum_{\sigma \in \mathfrak{S}_n} P_\sigma, \quad G := \frac{1}{n} \sum_{i=1}^n A_i,$$

so that $W_{\text{SS}} = \frac{1}{n!} \sum_{\sigma \in \mathfrak{S}_n} P_\sigma^K$, $W_{\text{RS}} = R^K$, and $W_{\text{GD}} = G^{nK}$.

Lemma 1.4 (Symmetry). *Let $A_1, \dots, A_n$ be real symmetric. For every $\sigma \in \mathfrak{S}_n$,*

$$P_\sigma^\top = A_{\sigma(n)} \cdots A_{\sigma(1)} = P_{\sigma^{\text{rev}}}, \quad \sigma^{\text{rev}}(j) := \sigma(n+1-j).$$

Consequently:

1. R and W_{SS} are symmetric, even though the individual products P_σ need not be;
2. G , $W_{\text{RS}} = R^K$, and $W_{\text{GD}} = G^{nK}$ are symmetric, and—an even power of a symmetric matrix being positive semidefinite— $W_{\text{RS}} \succeq 0$ when K is even and $W_{\text{GD}} \succeq 0$ when nK is even.

Proof. Since each $A_i = A_i^\top$, we have $P_\sigma^\top = (A_{\sigma(1)} \cdots A_{\sigma(n)})^\top = A_{\sigma(n)} \cdots A_{\sigma(1)} = P_{\sigma^{\text{rev}}}$. Reversal $\sigma \mapsto \sigma^{\text{rev}}$ is a bijection of $\mathfrak{S}_n$, so $R^\top = \frac{1}{n!} \sum_{\sigma} P_{\sigma^{\text{rev}}} = R$ and, using $(P_\sigma^K)^\top = (P_\sigma^\top)^K = P_{\sigma^{\text{rev}}}^K$, also $W_{\text{SS}}^\top = W_{\text{SS}}$. The average G is symmetric, and $W_{\text{RS}} = R^K$, $W_{\text{GD}} = G^{nK}$ are powers of symmetric matrices, hence symmetric. Finally, for symmetric M and $m \geq 1$, $M^{2m} = (M^m)^2 \succeq 0$. $\square$

1.2 Refutation of the SS–RS inequality

1.2.1 The construction

The starting point is the appendix observation of [10] that the first inequality already fails for rank-one 2×2 matrices at $\eta = 1$. We push that failure to the near-identity regime by a tensor square. Let $u_1, u_2, u_3 \in \mathbb{R}^2$ be the three unit vectors

$$u_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad u_2 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}, \quad u_3 = \frac{1}{2} \begin{pmatrix} -1 \\ \sqrt{3} \end{pmatrix},$$

and let $P_i = u_i u_i^\top$ be the corresponding rank-one orthogonal projectors:

$$P_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_2 = \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}, \quad P_3 = \begin{pmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}. \quad (1.3)$$

Each P_i is symmetric, idempotent, of trace 1. Fix a parameter $q \in (0, 1)$ and set

$$B_i := qI_2 + (1-q)P_i, \quad A_i := B_i \otimes B_i \in \mathbb{R}^{4 \times 4}. \quad (1.4)$$

Since P_i has eigenvalues 1 and 0, each B_i has eigenvalues 1 and q , so B_i is positive definite with $qI_2 \preceq B_i \preceq I_2$. Hence $A_i = B_i \otimes B_i$ is symmetric positive definite with eigenvalues $\{1, q, q, q^2\}$, giving

$$q^2 I_4 \preceq A_i \preceq I_4. \quad (1.5)$$

Given any proposed $\eta_{3,2} \in (0, 1]$, choose $q < 1$ close enough to 1 that $1 - q^2 < \eta_{3,2}$; then $(1 - \eta_{3,2})I_4 \preceq A_i \preceq I_4$, so the A_i satisfy the hypothesis of Conjecture 1.1.

1.2.2 Spectral norms of W_{SS} and W_{RS}

With $n = 3$ and $K = 2$ we compare

$$W_{\text{SS}} = \frac{1}{6} \sum_{\sigma \in \mathfrak{S}_3} P_{\sigma}^2, \quad W_{\text{RS}} = R^2$$

where

$$R = \frac{1}{6} \sum_{\sigma \in \mathfrak{S}_3} P_{\sigma}, \quad P_{\sigma} = A_{\sigma(1)} A_{\sigma(2)} A_{\sigma(3)}.$$

Both matrices are symmetric (Lemma 1.4); their eigenvalues are explicit polynomials in q , recorded in the next lemma. Every polynomial identity stated below is exact and is verified by the symbolic-algebra script of Section 1.4.

Lemma 1.5 (Eigenvalues). *The symmetric matrix W_{RS} has eigenvalues*

$$(\lambda_{\text{RS}}(q), \mu_{\text{RS}}(q), \mu_{\text{RS}}(q), q^6),$$

and the symmetric matrix W_{SS} has eigenvalues

$$(\lambda_{\text{SS}}(q), \mu_{\text{SS}}(q), \mu_{\text{SS}}(q), q^6),$$

where

$$\begin{aligned} \lambda_{\text{RS}}(q) &= \left(\frac{q^6 - 6q^5 + 15q^4 + 12q^3 + 15q^2 - 6q + 1}{32} \right)^2, \\ \mu_{\text{RS}}(q) &= \frac{(q+1)^4(q^4 + 4q^3 - 42q^2 + 4q + 1)^2}{16384}, \\ \lambda_{\text{SS}}(q) &= \frac{1}{2048} (q^{12} - 24q^{11} + 186q^{10} - 504q^9 + 399q^8 + 528q^7 + 876q^6 \\ &\quad + 528q^5 + 399q^4 - 504q^3 + 186q^2 - 24q + 1), \\ \mu_{\text{SS}}(q) &= -\frac{1}{8192} (q^{12} - 12q^{11} - 78q^{10} + 1188q^9 - 2001q^8 - 1176q^7 - 4036q^6 \\ &\quad - 1176q^5 - 2001q^4 + 1188q^3 - 78q^2 - 12q + 1). \end{aligned}$$

Moreover, for all $q \in (0, 1]$,

$$\lambda_{\text{RS}}(q) - q^6 = \frac{(1-q)^6(q+1)^2(q^4 - 8q^3 + 30q^2 - 8q + 1)}{1024} \geq 0, \quad (1.6)$$

$$\lambda_{\text{RS}}(q) - \mu_{\text{RS}}(q) = \frac{(1-q)^4(5q^2 + 2q + 5)(3q^6 - 30q^5 + 93q^4 + 124q^3 + 93q^2 - 30q + 3)}{16384} \geq 0. \quad (1.7)$$

In particular $\|W_{\text{RS}}\| = \lambda_{\text{RS}}(q)$ for every $q \in (0, 1]$.

Proof. Form $W_{\text{RS}} = R^2$ and $W_{\text{SS}} = \frac{1}{6} \sum_{\sigma \in \mathfrak{S}_3} P_{\sigma}^2$ as 4×4 matrices from $A_i = B_i \otimes B_i$ of (1.4). Each assertion of the lemma is an exact identity in q , verified by the symbolic-algebra computation of Section 1.4: the characteristic polynomials of W_{RS} and W_{SS} factor with exactly the stated eigenvalues and multiplicities, and the differences (1.6)–(1.7) equal the displayed expressions.

It remains to deduce $\|W_{\text{RS}}\| = \lambda_{\text{RS}}$. The cofactors in (1.6)–(1.7) are positive on $(0, 1]$ by the explicit sum-of-squares certificates

$$\begin{aligned} q^4 - 8q^3 + 30q^2 - 8q + 1 &= (q^2 - 4q + 1)^2 + 12q^2 \\ &\quad 5q^2 + 2q + 5 > 0, \\ 3q^6 - 30q^5 + 93q^4 + 124q^3 + 93q^2 - 30q + 3 \\ &= 3q^4((q - 5)^2 + 6) + 124q^3 + 3(31q^2 - 10q + 1) > 0 \quad (q \geq 0), \end{aligned}$$

and $(1 - q)^4, (1 - q)^6, (q + 1)^2, (q + 1)^4 \geq 0$. Hence the right-hand sides of (1.6)–(1.7) are nonnegative on $(0, 1]$. Since $W_{\text{RS}} = R^2 \succeq 0$ (Lemma 1.4), its norm is its largest eigenvalue, and these inequalities make λ_{RS} exceed both q^6 and μ_{RS} ; therefore $\|W_{\text{RS}}\| = \lambda_{\text{RS}}$. $\square$

Lemma 1.6 (The violation). *For all q ,*

$$\lambda_{\text{SS}}(q) - \lambda_{\text{RS}}(q) = \frac{(1 - q)^6(q + 1)^2(42q^2 - q^4 - 4q^3 - 4q - 1)}{2048}, \quad (1.8)$$

the final factor $g(q) := 42q^2 - q^4 - 4q^3 - 4q - 1$ has $g(q) > 0$ for $q \in (q^*, 1]$ for $q^* = 0.212036\dots$. Consequently $\lambda_{\text{SS}}(q) > \lambda_{\text{RS}}(q)$ for every $q \in (q^*, 1)$.

Proof. Subtracting the explicit eigenvalues λ_{SS} and λ_{RS} of Lemma 1.5 and expanding yields the polynomial identity (1.8). We determine the sign of its final factor $g(q) = 42q^2 - q^4 - 4q^3 - 4q - 1$ on $(0, 1]$. Since $g(0) = -1 < 0$ and $g(1) = 32 > 0$, $g''(q) = -12(q^2 + 2q - 7) > 0$ on $[0, 1]$, with $g(0) < 0 < g(1)$ this forces a single sign change, at some $q^* \in (0, 1)$, numerically $q^* = 0.212036\dots$. $\square$

1.2.3 Proof of Theorem 1.2

Fix any $q \in (q^*, 1)$. By Lemma 1.5, $\|W_{\text{RS}}\| = \lambda_{\text{RS}}(q)$. Since W_{SS} is symmetric (Lemma 1.4) and $\lambda_{\text{SS}}(q)$ is one of its eigenvalues, the spectral norm satisfies $\|W_{\text{SS}}\| \geq \lambda_{\text{SS}}(q)$. Combining this with Lemma 1.6,

$$\|W_{\text{SS}}\| \geq \lambda_{\text{SS}}(q) > \lambda_{\text{RS}}(q) = \|W_{\text{RS}}\|.$$

Now let $\eta_{3,2} \in (0, 1]$ be arbitrary. Choosing $q \in (q^*, 1)$ with $1 - q^2 < \eta_{3,2}$ (possible since $q^* < 1$), the matrices A_i of (1.4) are symmetric, positive definite, satisfy $(1 - \eta_{3,2})I \preceq A_i \preceq I$ by (1.5), and yet $\|W_{\text{SS}}\| > \|W_{\text{RS}}\|$. Therefore no $\eta_{3,2}$ validates the first inequality of (1.2), so Conjecture 1.1 fails at $(n, K) = (3, 2)$.

1.3 The RS–GD inequality holds

We now prove Theorem 1.3. This is the Recht–Ré AM–GM inequality [8] (raised to the K -th power), restricted to well-conditioned matrices. The unrestricted inequality is false for $n \geq 5$ [4, 2], but those counterexamples live far from I ; the theorem shows that a dimension-free conditioning radius $\eta = \frac{1}{4n^2 + 1}$ rules them out.

1.3.1 Reduction to one epoch

Since R is symmetric (Section 1.1) and $W_{\text{RS}} = R^K$, $W_{\text{GD}} = G^{nK}$ with $G \succ 0$ symmetric, we have $\|W_{\text{RS}}\| = \|R\|^K$ and $\|W_{\text{GD}}\| = \|G\|^{nK} = \rho^{nK}$ where $\rho := \|G\| = \lambda_{\max}(G)$. Hence it suffices to prove the single-epoch bound

$$\|R\| \leq \rho^n, \quad (1.9)$$

which is independent of K ; raising (1.9) to the K -th power (legitimate because $R \succeq 0$, established below) yields $\|W_{\text{RS}}\| = \|R\|^K \leq \rho^{nK} = \|W_{\text{GD}}\|$.

Because each A_i has eigenvalues in $[1 - \eta, 1]$, so does G , hence $1 - \eta \leq \rho \leq 1$. Normalize

$$C_i := \rho^{-1} A_i = I + X_i, \quad X_i := \rho^{-1} A_i - I, \quad \bar{C} := \frac{1}{n} \sum_i C_i = \rho^{-1} G.$$

Each $C_i \succ 0$ and each X_i is symmetric. Since $G \preceq \rho I$, we get $\bar{C} \preceq I$. The eigenvalues of X_i lie in $[\rho^{-1}(1 - \eta) - 1, \rho^{-1} - 1]$; using $1 - \eta \leq \rho \leq 1$, the upper end satisfies $\rho^{-1} - 1 \leq \frac{1}{1 - \eta} - 1 = \frac{\eta}{1 - \eta}$ and the lower end satisfies $|\rho^{-1}(1 - \eta) - 1| \leq \eta \leq \frac{\eta}{1 - \eta}$, so $\|X_i\| \leq \frac{\eta}{1 - \eta}$. With $\eta = \frac{1}{4n^2 + 1}$,

$$\delta := \max_i \|X_i\| \leq \frac{\eta}{1 - \eta} = \frac{1}{4n^2}. \quad (1.10)$$

Finally $\tilde{R} := \frac{1}{n!} \sum_{\sigma} \prod_i C_{\sigma(i)} = \rho^{-n} R$, so (1.9) is equivalent to $\tilde{R} \preceq I$ together with $\tilde{R} \succeq 0$. Both are supplied by the following lemma, applied to $C_i = \rho^{-1} A_i$.

1.3.2 A near-identity shuffled AM–GM lemma

Lemma 1.7 (Near-identity shuffled product). *Let $X_1, \dots, X_n$ be symmetric $d \times d$ matrices with $C_i := I + X_i \succeq 0$ and $\frac{1}{n} \sum_i C_i \preceq I$, and suppose $\delta := \max_i \|X_i\| \leq \frac{1}{4n^2}$. Then*

$$0 \preceq \tilde{R} := \frac{1}{n!} \sum_{\sigma \in \mathfrak{S}_n} \prod_{i=1}^n C_{\sigma(i)} \preceq I.$$

Proof. Symmetry and expansion. As $\mathfrak{S}_n$ is closed under reversal and reversing a product transposes it, $\tilde{R}^\top = \tilde{R}$. Expanding $\prod_i (I + X_{\sigma(i)})$ and averaging over σ ,

$$\tilde{R} = I + \sum_{k=1}^n E_k, \quad E_k = \frac{1}{k!} \sum_{\substack{(i_1, \dots, i_k) \\ \text{distinct}}} X_{i_1} X_{i_2} \cdots X_{i_k}. \quad (1.11)$$

Indeed, for fixed k each size- k subset of the n factor slots contributes, after averaging, the same term $\frac{(n-k)!}{n!} \sum_{\text{distinct}} X_{i_1} \cdots X_{i_k}$ (sum over distinct ordered k -tuples); there are $\binom{n}{k}$ such subsets, and $\binom{n}{k} \frac{(n-k)!}{n!} = \frac{1}{k!}$, giving (1.11).

Low-order terms. Put

$$T := - \sum_i X_i, \quad Q := \sum_i X_i^2 \succeq 0.$$

The hypothesis $\frac{1}{n} \sum_i C_i = I + \frac{1}{n} \sum_i X_i \preceq I$ is exactly $\sum_i X_i \preceq 0$, i.e. $T \succeq 0$. From (1.11),

$$E_1 = \sum_i X_i = -T, \quad E_2 = \frac{1}{2} \sum_{i \neq j} X_i X_j = \frac{1}{2} \left[\left(\sum_i X_i \right)^2 - \sum_i X_i^2 \right] = \frac{1}{2} (T^2 - Q).$$

Hence

$$I - \tilde{R} = T + \frac{1}{2}(Q - T^2) - \sum_{k=3}^n E_k. \quad (1.12)$$

Upper bound $\tilde{R} \preceq I$. Fix a unit vector v and write $t := v^\top T v \geq 0$ and $q := v^\top Q v = \sum_i \|X_i v\|^2 \geq 0$. Since $T \succeq 0$ and $\|T\| \leq \sum_i \|X_i\| \leq n\delta$, the operator inequality $T^2 \preceq \|T\| T$ gives $v^\top T^2 v \leq n\delta t$, so

$$v^\top \left(T - \frac{1}{2} T^2 \right) v \geq \left(1 - \frac{n\delta}{2} \right) t \geq 0, \quad (1.13)$$

using $n\delta \leq \frac{1}{4n} \leq \frac{1}{2}$. For the higher-order terms, each word with $k \geq 3$ obeys, by Cauchy–Schwarz on the two end factors and $\|X_{i_j}\| \leq \delta$ on the middle ones,

$$|v^\top X_{i_1} \cdots X_{i_k} v| \leq \|X_{i_1} v\| \delta^{k-2} \|X_{i_k} v\| \leq \frac{\delta^{k-2}}{2} (\|X_{i_1} v\|^2 + \|X_{i_k} v\|^2).$$

Summing over distinct ordered k -tuples: for each fixed first index there are $(n-1)(n-2)\cdots(n-k+1) =: (n-1)_{k-1}$ choices of the rest, and likewise for the last index, so $\sum_{\text{tuples}} \|X_{i_1} v\|^2 = (n-1)_{k-1} q$ and the same for the last factor. Therefore

$$|v^\top E_k v| \leq \frac{(n-1)_{k-1}}{k!} \delta^{k-2} q, \quad \sum_{k=3}^n |v^\top E_k v| \leq q \sum_{k=3}^n \frac{(n-1)_{k-1}}{k!} \delta^{k-2}. \quad (1.14)$$

Using $(n-1)_{k-1} \leq n^{k-1}$ and $\delta \leq \frac{1}{4n^2}$, the k -th summand is at most $\frac{1}{k!} n^{k-1} \delta^{k-2} = \frac{1}{k!} n (n\delta)^{k-2} \leq \frac{1}{k!} n (4n)^{-(k-2)}$; the $k=3$ term is $\leq \frac{1}{6} \cdot n \cdot \frac{1}{4n} = \frac{1}{24}$ and the series is dominated by a geometric tail, giving $\sum_{k \geq 3} \frac{(n-1)_{k-1}}{k!} \delta^{k-2} < \frac{1}{4}$ for all $n \geq 2$ (the supremum over n is $\frac{1}{24} + o(1) < \frac{1}{4}$). Combining (1.12), (1.13), and (1.14),

$$v^\top (I - \tilde{R}) v \geq \left(1 - \frac{n\delta}{2} \right) t + \frac{1}{2} q - \frac{1}{4} q \geq \frac{1}{4} q \geq 0,$$

so $\tilde{R} \preceq I$.

Lower bound $\tilde{R} \succeq 0$. From (1.11), each E_k is an average of at most $\frac{n!}{(n-k)!}$ words of norm $\leq \delta^k$, scaled by $\frac{1}{k!}$, so $\|E_k\| \leq \binom{n}{k} \delta^k$ and

$$\|\tilde{R} - I\| \leq \sum_{k=1}^n \binom{n}{k} \delta^k = (1 + \delta)^n - 1 \leq e^{n\delta} - 1 \leq e^{1/(4n)} - 1 < 1.$$

Thus every eigenvalue of $\tilde{R}$ exceeds $1 - (e^{1/(4n)} - 1) = 2 - e^{1/(4n)} > 0$, so $\tilde{R} \succ 0$. Combined with the upper bound, $0 \preceq \tilde{R} \preceq I$. $\square$

1.3.3 Proof of Theorem 1.3

Apply Lemma 1.7 to $C_i = \rho^{-1} A_i = I + X_i$: the hypotheses hold by the reduction above ($C_i \succ 0$, $\tilde{C} \preceq I$, and $\delta \leq \frac{1}{4n^2}$ by (1.10)). Since $\tilde{R} = \rho^{-n} R$, the lemma gives $0 \preceq R \preceq \rho^n I$, hence $R \succeq 0$ and $\|R\| \leq \rho^n$, which is (1.9). As $G \succeq 0$, $\|G^n\| = \rho^n$, so $\|R\| \leq \|G^n\|$. Finally, because $R \succeq 0$,

$$\|W_{RS}\| = \|R^K\| = \|R\|^K \leq \rho^{nK} = \|G^{nK}\| = \|W_{GD}\|.$$

This holds for every $n \geq 2$, $K \geq 1$, $d \geq 1$, and symmetric A_i with $(1 - \frac{1}{4n^2+1})I \preceq A_i \preceq I$. $\square$

Remark 1.8 (On the constant). The constant $\eta_{n,K}^{\text{RS-GD}} = \frac{1}{4n^2+1} = \Theta(1/n^2)$ is not optimized; the proof only uses $\delta \leq \frac{1}{4n^2}$ with slack (the tail in (1.14) is $< \frac{1}{24}$, not just $< \frac{1}{4}$). Notably it is independent of the epoch count K , matching the paper’s intuition that the reshuffling-versus-GD comparison should need only a mild, K -free conditioning. By contrast, Theorem 1.2 shows that for the single-shuffle inequality *no* positive constant works at $(n, K) = (3, 2)$, matrix-free.

1.4 Reproducible symbolic verification

All claims of Section 1.2 are exact polynomial identities in q . The following self-contained `sympy` script forms the matrices of the construction (1.3)–(1.4) and verifies them; it runs to completion, printing all identities verified with no failed assertion.

```
import sympy as sp, itertools
q, t = sp.symbols('q t')
s3 = sp.sqrt(3)
P1 = sp.Matrix([[1,0],[0,0]])
P2 = sp.Matrix([[sp.Rational(1,4), s3/4],[s3/4, sp.Rational(3,4)]])
P3 = sp.Matrix([[sp.Rational(1,4), -s3/4],[-s3/4, sp.Rational(3,4)]])
B = [q*sp.eye(2) + (1-q)*P for P in (P1, P2, P3)]
A = [sp.Matrix(sp.kronecker_product(b, b)) for b in B]
prm = list(itertools.permutations(range(3)))
R = sum((A[a]*A[b]*A[c] for a,b,c in prm), sp.zeros(4)) / 6
Wrs = sp.expand(R*R)
Wss = sp.expand(sum(((A[a]*A[b]*A[c])**2 for a,b,c in prm),
                    sp.zeros(4)) / 6)
assert sp.expand(Wrs - Wrs.T) == sp.zeros(4) # W_RS symmetric
assert sp.expand(Wss - Wss.T) == sp.zeros(4) # W_SS symmetric

# eigenvalue polynomials claimed in Lemma 5
lamRS = ((q**6-6*q**5+15*q**4+12*q**3+15*q**2-6*q+1)/32)**5
muRS = (q+1)**4*(q**4+4*q**3-42*q**2+4*q+1)**2/16384
lamSS = (q**12-24*q**11+186*q**10-504*q**9+399*q**8+528*q**7
        +876*q**6+528*q**5+399*q**4-504*q**3+186*q**2
        -24*q+1)/2048
muSS = -(q**12-12*q**11-78*q**10+1188*q**9-2001*q**8
        -1176*q**7-4036*q**6-1176*q**5-2001*q**4+1188*q**3
        -78*q**2-12*q+1)/8192

# characteristic polynomials factor with these roots
cRS = (t-lamRS)*(t-muRS)**2*(t-q**6)
cSS = (t-lamSS)*(t-muSS)**2*(t-q**6)
assert sp.factor(Wrs.charpoly(t).as_expr() - cRS) == 0
assert sp.factor(Wss.charpoly(t).as_expr() - cSS) == 0

# gap identity (Lemma 6) and dominance identities (Lemma 5)
g = 42*q**2-q**4-4*q**3-4*q-1
def zero(e): return sp.expand(e) == 0
assert zero(lamSS-lamRS - (1-q)**6*(q+1)**2*g/2048)
assert zero(lamRS-q**6
            - (1-q)**6*(q+1)**2*(q**4-8*q**3+30*q**2-8*q+1)/1024)
assert zero(lamRS-muRS - (1-q)**4*(5*q**2+2*q+5)
            *(3*q**6-30*q**5+93*q**4+124*q**3+93*q**2-30*q+3)/16384)

# sum-of-squares certificates from the proof of Lemma 5:
# positivity of the cofactors above, whence ||W_RS|| = lamRS
```

```

assert zero(q**4-8*q**3+30*q**2-8*q+1
            - ((q**2-4*q+1)**2 + 12*q**2))
assert zero(3*q**6-30*q**5+93*q**4+124*q**3+93*q**2-30*q+3
            - (3*q**4*((q-5)**2+6) + 124*q**3
              + 3*(31*q**2-10*q+1)))
print("all identities verified")

```

The script confirms that W_{RS} and W_{SS} are symmetric; that their characteristic polynomials factor as $(t - \lambda_{RS})(t - \mu_{RS})^2(t - q^6)$ and $(t - \lambda_{SS})(t - \mu_{SS})^2(t - q^6)$, giving the eigenvalue lists of Lemma 1.5; that the gap identity (1.8) and the dominance identities (1.6)–(1.7) hold; and that the two sum-of-squares certificates in the proof of Lemma 1.5 (which yield $\|W_{RS}\| = \lambda_{RS}$) are exact. Being exact symbolic identities in q , these are proofs of the corresponding statements.

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